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$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{x \cos x - \sin x}{e^x - e^{-x} - 2x} \\ &= \frac{0}{0} \stackrel{H}{=} \lim_{x \rightarrow 0} \frac{-x \sin x}{e^x + e^{-x} - 2} \\ &= \frac{0}{0} \stackrel{H}{=} \lim_{x \rightarrow 0} \frac{-\sin x - x \cos x}{e^x - e^{-x}} \\ &= \frac{0}{0} \stackrel{H}{=} \lim_{x \rightarrow 0} \frac{-2 \cos x + x \sin x}{e^x + e^{-x}} \\ &= -\frac{2}{2} = -1 \end{aligned}$$

References